Test Case Document

VendorSetu

Street Vendor Formalization & Micro-Credit Platform

30 test cases (5 modules x 6) in the format of the Test Case Template (Doc T_01)

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1. Test Plan and Results Summary

Item	Details
Objective	Verify that the four implemented modules of VendorSetu meet their functional requirements (SRS 4.1-4.6) and reject invalid input with the correct message.
Test cases	6 test cases for each of the 5 tested modules (30 in total, 162 steps), written in the format of the course Test Case Template (Doc T_01): header, pre-conditions, steps with the expected system response, Pass/Fail, comment, and post-conditions. Like the template's Example 2, each test case tests the happy path and also its failure conditions.
Design techniques (Comment column)	Positive (valid input); Negative / equivalence partitioning (EP) - every invalid class; boundary-value analysis (BVA) - both sides of each limit; Combination - several wrong fields at once; Security - forged requests, role checks, SQL injection; 'Check post-condition' steps look inside the database.
Levels	System testing through the web interface (24 test cases): the Actual Output is the message or value shown on screen. Unit testing of the Trust Score formula (TC-TS-02 to TC-TS-06).
Test runs	Run 1 - the first version of the code (git commit 0ce1911): 7 test cases FAILED; 6 defects were logged (Section 7). Run 2 - after fixing them, all test cases were executed again (regression testing). On the test case pages (Sections 2-6) every step that FAILED in Run 1 is marked 'Fail (Run 1)' in red and shows both outputs: the wrong Run-1 output and the correct output after the fix.
Environment / tools	Windows 11, Python 3.11, Flask 3.1, SQLite; pytest and the Flask test client. tests/run_test_cases.py performs every step on a fresh database and copies the on-screen result into the Actual Output column - nothing in that column is typed by hand.

Module (section)	Test cases	Cases	Steps	Run 1: cases failed	Run 2: cases passed	Run 2: steps passed
2. User Registration & Login	TC-AUTH-01 to TC-AUTH-06	6	37	1	6 / 6	37 / 37
3. Vendor Profile & Daily Sales	TC-SALE-01 to TC-SALE-06	6	36	2	6 / 6	36 / 36
4. Trust Score Calculation	TC-TS-01 to TC-TS-06	6	21	0	6 / 6	21 / 21
5. Loan Application & Bank Review	TC-LOAN-01 to TC-LOAN-06	6	35	2	6 / 6	35 / 35
6. EMI Schedule & Repayment	TC-EMI-01 to TC-EMI-06	6	33	2	6 / 6	33 / 33
Total		**30**	**162**	**7**	**30 / 30**	**162 / 162**

Automated run (pytest): 39 passed in 69.07s (0:01:09) - the 30 test cases, the combination suites and 2 integration tests. Executed on 04-11-2026.

Repeating a step by hand in the demo: the demo data (python seed_demo.py) already contains loans, so loan numbers such as #1 will differ; the messages are otherwise identical.

2. User Registration & Login (TC-AUTH-01 to TC-AUTH-06)

Test Case #:	TC-AUTH-01	**Test Case Name:**	Register a new account	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-1, FR-AUTH-2, FR-AUTH-4, NFR-SEC-1
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that a new user can register, that the password is stored as a hash and that a mobile number can be registered only once.				

Pre-conditions
The application is running with an empty database
The user is on the Register page

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Fill the Register form and click 'Create account'	Name = Ramesh Kumar, Mobile = 9876543210, Role = Street Vendor, Password = ramesh123, Confirm = ramesh123	Registration successful. Please log in. -> Log in page is shown	Registration successful. Please log in. -> Log in page is shown	**Pass**	Positive (happy path)
2	**Check post-condition 1**	-	1 user stored; password saved as a hash, not as 'ramesh123'	1 user stored; password saved as a hash, not as 'ramesh123'	**Pass**	
3	Register again with the same mobile number	Name = Suresh Kumar, Mobile = 9876543210, Password = suresh123	Mobile number already registered	Mobile number already registered	**Pass**	Negative - duplicate mobile
4	Register a bank officer with a new mobile number	Name = Anil Sharma, Mobile = 9123456780, Role = Bank / NBFC Officer, Password = bank123	Registration successful. Please log in. -> Log in page is shown	Registration successful. Please log in. -> Log in page is shown	**Pass**	Positive - another role
5	**Check post-condition 2**	-	2 users stored: Ramesh Kumar (VENDOR), Anil Sharma (BANK_OFFICER)	2 users stored: Ramesh Kumar (VENDOR), Anil Sharma (BANK_OFFICER)	**Pass**	

Post-conditions
1. Ramesh's password is stored only as a salted hash
2. Only the two valid registrations are stored; the duplicate mobile number is rejected

Test Case #:	TC-AUTH-02	**Test Case Name:**	Registration - name validation	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-1
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test the name rules: required, only letters, spaces and dots, at most 50 characters.				

Pre-conditions
The application is running with an empty database
The user is on the Register page; Role = Street Vendor and Password = ramesh123 in every step, and a new mobile number in every step

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Fill the Register form and click 'Create account'	Name = (empty), Mobile = 9876543210	Name is required	Name is required	**Pass**	Negative (EP) - empty
2	Fill the Register form and click 'Create account'	Name = 12345, Mobile = 9876543211	Name can contain only letters, spaces and dots	**Run 1:** Registration successful. Please log in. **Run 2 (after fix):** Name can contain only letters, spaces and dots	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - digits Run 1 FAILED: defect DEF-04 logged; fixed and re-tested
3	Fill the Register form and click 'Create account'	Name = 'A' x 51 characters, Mobile = 9876543212	Name must be at most 50 characters	**Run 1:** Registration successful. Please log in. **Run 2 (after fix):** Name must be at most 50 characters	**Fail (Run 1)** Pass (Run 2)	BVA - just above the limit Run 1 FAILED: defect DEF-04 logged; fixed and re-tested
4	Fill the Register form and click 'Create account'	Name = 'A' x 50 characters, Mobile = 9876543213	Registration successful. Please log in.	Registration successful. Please log in.	**Pass**	BVA - on the limit
5	Fill the Register form and click 'Create account'	Name = Mohd. Salim, Mobile = 9876543214	Registration successful. Please log in.	Registration successful. Please log in.	**Pass**	Positive - name with a dot
6	**Check post-condition 1**	-	2 users stored: 'A' x 50 (VENDOR), Mohd. Salim (VENDOR)	**Run 1:** 4 users stored: 12345 (VENDOR), 'A' x 51 (VENDOR), 'A' x 50 (VENDOR), Mohd. Salim (VENDOR) **Run 2 (after fix):** 2 users stored: 'A' x 50 (VENDOR), Mohd. Salim (VENDOR)	**Fail (Run 1)** Pass (Run 2)	Run 1 FAILED: wrong data stored (defect in the steps above); fixed and re-tested

Post-conditions
1. Only the two valid names are stored; every rejected name stores nothing

Test Case #:	TC-AUTH-03	**Test Case Name:**	Registration - mobile number and role validation	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-1, FR-AUTH-2
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that only a 10-digit Indian mobile number starting with 6-9 and one of the four roles are accepted.				

Pre-conditions
The application is running with an empty database
The user is on the Register page; Name = Ramesh Kumar and Password = ramesh123 in every step

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Fill the Register form and click 'Create account'	Mobile = 987654321 (9 digits)	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	**Pass**	BVA - 9 digits
2	Fill the Register form and click 'Create account'	Mobile = 98765432101 (11 digits)	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	**Pass**	BVA - 11 digits
3	Fill the Register form and click 'Create account'	Mobile = 5876543210 (starts with 5)	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	**Pass**	Negative (EP) - must start with 6-9
4	Fill the Register form and click 'Create account'	Mobile = 98765abcde	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	**Pass**	Negative (EP) - letters
5	Fill the Register form and click 'Create account'	Mobile = 9876543210, Role = (not selected)	Invalid role selected	Invalid role selected	**Pass**	Negative - no role
6	Send the form with a role that is not offered	Mobile = 9876543210, Role = ADMIN	Invalid role selected	Invalid role selected	**Pass**	Security - forged request
7	**Check post-condition 1**	-	0 users stored	0 users stored	**Pass**	

Post-conditions
1. No account is created by any of these inputs

Test Case #:	TC-AUTH-04	**Test Case Name:**	Registration - password validation	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-1, FR-AUTH-3
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the password rules (at least 6 characters with a letter and a digit, confirmation must match) and the message shown when every field is wrong.				

Pre-conditions

The application is running with an empty database

The user is on the Register page; Name = Ramesh Kumar, Mobile = 9876543210, Role = Street Vendor unless the step says otherwise

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Fill the Register form and click 'Create account'	Password = Confirm = ab123 (5 characters)	Password must be at least 6 characters with letters and digits	Password must be at least 6 characters with letters and digits	**Pass**	BVA - 5 characters
2	Fill the Register form and click 'Create account'	Password = Confirm = abcdefgh	Password must be at least 6 characters with letters and digits	Password must be at least 6 characters with letters and digits	**Pass**	Negative (EP) - no digit
3	Fill the Register form and click 'Create account'	Password = Confirm = 12345678	Password must be at least 6 characters with letters and digits	Password must be at least 6 characters with letters and digits	**Pass**	Negative (EP) - no letter
4	Fill the Register form and click 'Create account'	Password = ramesh123, Confirm = ramesh124	Passwords do not match	Passwords do not match	**Pass**	Negative - confirmation differs
5	Leave every field empty and click 'Create account'	All fields empty	Name is required	Name is required	**Pass**	Combination - first broken rule is reported
6	Fill the Register form and click 'Create account'	Password = Confirm = ab1234 (6 characters)	Registration successful. Please log in.	Registration successful. Please log in.	**Pass**	BVA - 6 characters
7	**Check post-condition 1**	-	1 user stored: Ramesh Kumar (VENDOR)	1 user stored: Ramesh Kumar (VENDOR)	**Pass**	

Post-conditions

1. Only the registration with a valid password (step 6) is stored

Test Case #:	TC-AUTH-05	**Test Case Name:**	Login and logout	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-5, FR-AUTH-7, NFR-SEC-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test login with correct and wrong credentials, logout, and that pages need a logged-in user.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered
The user is on the Log in page

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Enter mobile number and password; click 'Log in'	9876543210 / ramesh123	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	**Pass**	Positive (happy path)
2	Click 'Logout'	-	Logged out. -> Log in page is shown	Logged out. -> Log in page is shown	**Pass**	Positive
3	Log in with a wrong password	9876543210 / ramesh124	Invalid mobile number or password -> Log in page is shown	Invalid mobile number or password -> Log in page is shown	**Pass**	Negative - wrong password
4	Log in with a mobile number that is not registered	9000000001 / ramesh123	Invalid mobile number or password -> Log in page is shown	Invalid mobile number or password -> Log in page is shown	**Pass**	Negative - unknown user (same message, so no hint is given)
5	Type an SQL-injection string in both fields	' OR '1'='1 / ' OR '1'='1	Invalid mobile number or password -> Log in page is shown	Invalid mobile number or password -> Log in page is shown	**Pass**	Security - SQL injection
6	Type the dashboard address in the browser without logging in	URL = /dashboard	Please log in first. -> Log in page is shown	Please log in first. -> Log in page is shown	**Pass**	Security - page needs login

Post-conditions
1. A session exists only after step 1; after a failed login the user stays on the Log in page

Test Case #:	TC-AUTH-06	**Test Case Name:**	Account lock after 5 failed attempts	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-6, NFR-SEC-2
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that 5 consecutive wrong passwords lock the account and that a successful login resets the counter.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and not locked
The user is on the Log in page

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Log in with a wrong password 4 times	9876543210 / wrong1 (x4)	Invalid mobile number or password (each time)	Invalid mobile number or password (each time)	**Pass**	Negative
2	Log in with the correct password	9876543210 / ramesh123	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	**Pass**	Successful login resets the counter
3	Log in with a wrong password 4 more times	9876543210 / wrong1 (x4)	Invalid mobile number or password (each time)	Invalid mobile number or password (each time)	**Pass**	Not locked: the counter was reset
4	Log in with a wrong password a 5th time	9876543210 / wrong1	Account locked after 5 failed attempts. Contact support	Account locked after 5 failed attempts. Contact support	**Pass**	BVA - 5th attempt locks
5	Log in with the correct password	9876543210 / ramesh123	Account locked after 5 failed attempts. Contact support	Account locked after 5 failed attempts. Contact support	**Pass**	Locked even with the correct password
6	**Check post-condition 1**	-	is_locked = 1, failed_attempts = 5	is_locked = 1, failed_attempts = 5	**Pass**	

Post-conditions
1. Ramesh's account is locked (is_locked = 1) after 5 consecutive failed attempts

3. Vendor Profile & Daily Sales (TC-SALE-01 to TC-SALE-06)

Test Case #:	TC-SALE-01	**Test Case Name:**	Save the profile and log a daily sale	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-VEN-1, FR-VEN-3, FR-VEN-5, FR-VEN-6
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that sales can be logged only after the profile is saved, and only once per day.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
He has NOT saved his business profile yet and has no sales

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026, Amount = 1500, Mode = UPI	Complete your vendor profile first	Complete your vendor profile first	**Pass**	Negative - profile missing
2	In 'Business profile' enter the values and click 'Save profile'	Business = Fruits & vegetables, Zone = Sector 14 Market, Gurugram, Years = 5, UPI = ramesh@upi	Profile saved.	Profile saved.	**Pass**	Positive
3	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026, Amount = 1500, Mode = UPI	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	Positive (happy path)
4	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026 (same day), Amount = 900, Mode = Cash	Sales already logged for this date	Sales already logged for this date	**Pass**	Negative - one entry per day
5	**Check post-condition 1**	-	1 sale stored: 04-11-2026 Rs 1500 UPI	1 sale stored: 04-11-2026 Rs 1500 UPI	**Pass**	

Post-conditions
1. The profile is saved and exactly one sale is stored for today; the duplicate entry is rejected

Test Case #:	TC-SALE-02	**Test Case Name:**	Business profile validation	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-VEN-1, FR-VEN-2
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that business type and vending zone are required and years of vending must be 0-60.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
He has NOT saved his business profile; in every step the other fields are valid (Business = Fruits & vegetables, Zone = Sector 14 Market, Gurugram, Years = 5)

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Save the profile	Business type = (empty)	Business type and vending zone are required	Business type and vending zone are required	**Pass**	Negative - required field
2	Save the profile	Vending zone = (empty)	Business type and vending zone are required	Business type and vending zone are required	**Pass**	Negative - required field
3	Save the profile	Years = abc	Years of vending must be a number	Years of vending must be a number	**Pass**	Negative (EP) - not a number
4	Save the profile	Years = 61	Years of vending must be between 0 and 60	Years of vending must be between 0 and 60	**Pass**	BVA - just above 60
5	Save the profile	Years = -1	Years of vending must be between 0 and 60	Years of vending must be between 0 and 60	**Pass**	BVA - just below 0
6	Save the profile	Years = 60	Profile saved.	Profile saved.	**Pass**	BVA - on the limit
7	**Check post-condition 1**	-	Profile stored: Fruits & vegetables, Sector 14 Market, Gurugram, 60 years	Profile stored: Fruits & vegetables, Sector 14 Market, Gurugram, 60 years	**Pass**	

Post-conditions
1. Only the valid profile (step 6) is stored

Test Case #:	TC-SALE-03	**Test Case Name:**	Sale amount - invalid values	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-VEN-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test that zero, negative, non-numeric and special values (nan, inf) are rejected as the amount.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His profile is saved and he has no sales; Date = today and Mode = UPI in every step

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	In 'Daily sales' enter the values and click 'Log sale'	Amount = 0	Sale amount must be greater than 0	Sale amount must be greater than 0	**Pass**	BVA - zero
2	In 'Daily sales' enter the values and click 'Log sale'	Amount = -500	Sale amount must be greater than 0	Sale amount must be greater than 0	**Pass**	Negative (EP) - negative
3	In 'Daily sales' enter the values and click 'Log sale'	Amount = abc	Sale amount must be a number	Sale amount must be a number	**Pass**	Negative (EP) - text
4	In 'Daily sales' enter the values and click 'Log sale'	Amount = (empty)	Sale amount must be a number	Sale amount must be a number	**Pass**	Negative (EP) - empty
5	In 'Daily sales' enter the values and click 'Log sale'	Amount = nan	Sale amount must be a number	**Run 1:** HTTP 500 Internal Server Error (IntegrityError: NOT NULL constraint failed: sales.amount) **Run 2 (after fix):** Sale amount must be a number	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - special value Run 1 FAILED: defect DEF-01 logged; fixed and re-tested
6	In 'Daily sales' enter the values and click 'Log sale'	Amount = inf	Sale amount must be a number	**Run 1:** Sale amount cannot exceed Rs 1,00,000 per day **Run 2 (after fix):** Sale amount must be a number	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - special value Run 1 FAILED: defect DEF-01 logged; fixed and re-tested
7	**Check post-condition 1**	-	0 sales stored	0 sales stored	**Pass**	

Post-conditions
1. No sale is stored for any invalid amount

Test Case #:	TC-SALE-04	**Test Case Name:**	Sale amount - boundary values	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-VEN-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the amounts on both sides of the limits: greater than 0 and at most Rs 1,00,000 per day.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His profile is saved and he has no sales; Mode = UPI in every step

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026, Amount = 100000.01	Sale amount cannot exceed Rs 1,00,000 per day	Sale amount cannot exceed Rs 1,00,000 per day	**Pass**	BVA - just above the upper limit
2	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026, Amount = 100000	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	BVA - on the upper limit
3	In 'Daily sales' enter the values and click 'Log sale'	Date = 03-11-2026, Amount = 1	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	BVA - smallest whole amount
4	In 'Daily sales' enter the values and click 'Log sale'	Date = 02-11-2026, Amount = -0.01	Sale amount must be greater than 0	Sale amount must be greater than 0	**Pass**	BVA - just below the lower limit
5	**Check post-condition 1**	-	2 sales stored: 04-11-2026 Rs 100000 UPI, 03-11-2026 Rs 1 UPI	2 sales stored: 04-11-2026 Rs 100000 UPI, 03-11-2026 Rs 1 UPI	**Pass**	

Post-conditions
1. Only the two amounts inside the limits are stored

Test Case #:	TC-SALE-05	**Test Case Name:**	Sale date and payment mode validation	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-VEN-3, FR-VEN-4, FR-VEN-5
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test the date rules (not in the future, not more than 90 days old, a real date), the payment mode and an entry with several wrong fields.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His profile is saved and he has no sales; Amount = 1500 and Mode = UPI unless the step says otherwise

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	In 'Daily sales' enter the values and click 'Log sale'	Date = 05-11-2026 (tomorrow)	Sale date cannot be in the future	Sale date cannot be in the future	**Pass**	BVA - tomorrow
2	In 'Daily sales' enter the values and click 'Log sale'	Date = 2026-02-30 (no such day)	Enter a valid date (YYYY-MM-DD)	Enter a valid date (YYYY-MM-DD)	**Pass**	Negative (EP) - invalid date
3	In 'Daily sales' enter the values and click 'Log sale'	Date = 05-08-2026 (91 days ago)	Sale date cannot be more than 90 days old	**Run 1:** Sale logged. Your trust score has been updated. **Run 2 (after fix):** Sale date cannot be more than 90 days old	**Fail (Run 1)** Pass (Run 2)	BVA - just outside 90 days Run 1 FAILED: defect DEF-02 logged; fixed and re-tested
4	In 'Daily sales' enter the values and click 'Log sale'	Date = 06-08-2026 (90 days ago)	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	BVA - on the 90-day limit
5	Send the sales form with a payment mode that is not offered	Date = 04-11-2026, Mode = CARD	Payment mode must be CASH or UPI	Payment mode must be CASH or UPI	**Pass**	Security - forged value
6	In 'Daily sales' enter the values and click 'Log sale'	Amount = 0, Date = 05-11-2026, Mode = CARD (three fields wrong)	Sale amount must be greater than 0	Sale amount must be greater than 0	**Pass**	Combination - first broken rule is reported
7	**Check post-condition 1**	-	1 sale stored: 06-08-2026 Rs 1500 UPI	**Run 1:** 2 sales stored: 06-08-2026 Rs 1500 UPI, 05-08-2026 Rs 1500 UPI **Run 2 (after fix):** 1 sale stored: 06-08-2026 Rs 1500 UPI	**Fail (Run 1)** Pass (Run 2)	Run 1 FAILED: wrong data stored (defect in the steps above); fixed and re-tested

Post-conditions
1. Only the entry dated 90 days ago (step 4) is stored

Test Case #:	TC-SALE-06	**Test Case Name:**	30-day sales summary on the dashboard	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-VEN-7, FR-TS-1
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that the dashboard shows the correct 30-day total, active days, daily average and the updated Trust Score.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His profile is saved (5 years of vending) and he has no sales

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	In 'Daily sales' enter the values and click 'Log sale'	Date = 02-11-2026, Amount = 1000, Mode = Cash	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	Positive
2	In 'Daily sales' enter the values and click 'Log sale'	Date = 03-11-2026, Amount = 2000, Mode = Cash	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	Positive
3	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026, Amount = 3000, Mode = Cash	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	Positive
4	Read the summary cards on the dashboard	-	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	**Pass**	Hand calculation: 6000 / 3 = 2000
5	Read the Trust Score card	-	Trust Score 34/100, HIGH RISK	Trust Score 34/100, HIGH RISK	**Pass**	Hand calculation: 3/30 x 40 + 20 + 0 + 10 = 34

Post-conditions
1. Three sales are stored and the summary and Trust Score reflect them

4. Trust Score Calculation (TC-TS-01 to TC-TS-06)

Test Case #:	TC-TS-01	**Test Case Name:**	Trust Score of a new vendor on the dashboard	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-1, FR-TS-3, FR-TS-4, FR-TS-5, FR-TS-6
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the score, risk band and component points shown to a new vendor, and that the score is updated and used for loan eligibility.				

Pre-conditions
Vendor Sunita Devi (9812345670 / sunita123) is registered; her profile shows 0 years of vending; she has no sales, vouches or loans

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Log in as Sunita and read the Trust Score card	9812345670 / sunita123	Trust Score 20/100, HIGH RISK	Trust Score 20/100, HIGH RISK	**Pass**	Hand calculation: 0 + 20 (no EMI due) + 0 + 0
2	Read the four component bars	-	Sales consistency 0/40, Repayment history 20/40, Peer vouching 0/10, Vending tenure 0/10	Sales consistency 0/40, Repayment history 20/40, Peer vouching 0/10, Vending tenure 0/10	**Pass**	Explainable score
3	In 'Daily sales' enter the values and click 'Log sale'	Date = 04-11-2026, Amount = 800, Mode = Cash	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**Pass**	Positive
4	Read the Trust Score card again	-	Trust Score 21/100, HIGH RISK	Trust Score 21/100, HIGH RISK	**Pass**	1/30 x 40 = 1.3; 1.3 + 20 = 21
5	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 10000	Trust score 21 is below 40. Not eligible	Trust score 21 is below 40. Not eligible	**Pass**	Negative - score below 40

Post-conditions
1. One sale is stored; no loan is created

Test Case #:	TC-TS-02	**Test Case Name:**	Formula - loan eligibility boundary (39 / 40)	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)	**Type:**	Unit test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-2, FR-TS-4, FR-TS-5
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the score on both sides of the eligibility cut-off of 40.				

Pre-conditions
compute_trust_score() is called directly (unit test)
Window = 30 days; no EMI due yet (repayment = 20); Vouches = 0; Years = 0

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Call compute_trust_score() with the input	Active days = 14	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	**Pass**	BVA - just below 40
2	Call compute_trust_score() with the input	Active days = 15	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	**Pass**	BVA - on 40
3	Call compute_trust_score() with the input	Active days = 16	21.3 + 20 + 0 + 0 = 41 -> MEDIUM RISK, eligible	21.3 + 20 + 0 + 0 = 41 -> MEDIUM RISK, eligible	**Pass**	BVA - just above 40

Post-conditions
1. Score, band and eligibility are returned; no stored data changes

Test Case #:	TC-TS-03	**Test Case Name:**	Formula - risk band boundary (69 / 70)	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)	**Type:**	Unit test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-2, FR-TS-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the score on both sides of the MEDIUM / LOW RISK boundary.				

Pre-conditions
compute_trust_score() is called directly (unit test)
Window = 30 days; no EMI due yet (repayment = 20); Years = 5

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Call compute_trust_score() with the input	Active days = 29, Vouches = 0	38.7 + 20 + 0 + 10 = 69 -> MEDIUM RISK, eligible	38.7 + 20 + 0 + 10 = 69 -> MEDIUM RISK, eligible	**Pass**	BVA - just below 70
2	Call compute_trust_score() with the input	Active days = 30, Vouches = 0	40 + 20 + 0 + 10 = 70 -> LOW RISK, eligible	40 + 20 + 0 + 10 = 70 -> LOW RISK, eligible	**Pass**	BVA - on 70
3	Call compute_trust_score() with the input	Active days = 28, Vouches = 1	37.3 + 20 + 3.3 + 10 = 71 -> LOW RISK, eligible	37.3 + 20 + 3.3 + 10 = 71 -> LOW RISK, eligible	**Pass**	BVA - just above 70

Post-conditions
1. Score, band and eligibility are returned; no stored data changes

Test Case #:	TC-TS-04	**Test Case Name:**	Formula - minimum, maximum and repayment history	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)	**Type:**	Unit test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-2, FR-TS-3, FR-TS-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the lowest and highest possible scores and the effect of the repayment record.				

Pre-conditions
compute_trust_score() is called directly (unit test)
Window = 30 days

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Call compute_trust_score() with the input	Active days = 0, EMIs due = 0, Vouches = 0, Years = 0	0 + 20 + 0 + 0 = 20 -> HIGH RISK, not eligible	0 + 20 + 0 + 0 = 20 -> HIGH RISK, not eligible	**Pass**	BVA - minimum
2	Call compute_trust_score() with the input	Active days = 30, On-time EMIs = 12 of 12, Vouches = 3, Years = 5	40 + 40 + 10 + 10 = 100 -> LOW RISK, eligible	40 + 40 + 10 + 10 = 100 -> LOW RISK, eligible	**Pass**	BVA - maximum
3	Call compute_trust_score() with the input	Active days = 30, On-time EMIs = 0 of 6, Vouches = 0, Years = 5	40 + 0 + 0 + 10 = 50 -> MEDIUM RISK, eligible	40 + 0 + 0 + 10 = 50 -> MEDIUM RISK, eligible	**Pass**	No EMI paid on time
4	Call compute_trust_score() with the input	Active days = 24, On-time EMIs = 6 of 12, Vouches = 1, Years = 2	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	**Pass**	Positive - partial record

Post-conditions
1. Score, band and eligibility are returned; no stored data changes

Test Case #:	TC-TS-05	**Test Case Name:**	Formula - caps on vouches and tenure	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)	**Type:**	Unit test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-2, BR-8
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that only 3 vouches and 5 years of vending count toward the score.				

Pre-conditions
compute_trust_score() is called directly (unit test)
Active days = 30 of 30; no EMI due yet (repayment = 20)

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Call compute_trust_score() with the input	Vouches = 3, Years = 5	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	**Pass**	BVA - on the caps
2	Call compute_trust_score() with the input	Vouches = 7, Years = 20	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	**Pass**	BVA - above the caps (same score)
3	Call compute_trust_score() with the input	Vouches = 2, Years = 4	40 + 20 + 6.7 + 8 = 75 -> LOW RISK, eligible	40 + 20 + 6.7 + 8 = 75 -> LOW RISK, eligible	**Pass**	BVA - below the caps

Post-conditions
1. Score, band and eligibility are returned; no stored data changes

Test Case #:	TC-TS-06	**Test Case Name:**	Formula - invalid inputs	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)	**Type:**	Unit test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-1
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that impossible input values are rejected with a clear message.				

Pre-conditions
compute_trust_score() is called directly (unit test)

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Call compute_trust_score() with the input	Active days = 35, Window = 30	Active days cannot exceed the observation window	Active days cannot exceed the observation window	**Pass**	Negative - more days than the window
2	Call compute_trust_score() with the input	Active days = 30, Vouches = -1, Years = 5	Score inputs cannot be negative	Score inputs cannot be negative	**Pass**	Negative - negative value
3	Call compute_trust_score() with the input	Active days = 30, On-time EMIs = 5, EMIs due = 3	On-time EMIs cannot exceed EMIs due	On-time EMIs cannot exceed EMIs due	**Pass**	Negative - inconsistent values

Post-conditions
1. No score is returned for invalid input

5. Loan Application & Bank Review (TC-LOAN-01 to TC-LOAN-06)

Test Case #:	TC-LOAN-01	**Test Case Name:**	Eligible vendor applies for a Tier-1 loan	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-LN-1, FR-LN-2, FR-LN-4, FR-LN-5, FR-LN-6
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test a loan application at and above the Tier-1 limit and the one-active-loan rule.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His profile shows 5 years of vending and he logged a sale on each of the last 30 days, so his Trust Score is 40 + 20 + 0 + 10 = 70 (LOW RISK)
He has no loan

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Read the Trust Score and the eligible-limit card on the dashboard	-	Trust Score 70/100, LOW RISK; Eligible limit Rs 15,000 (Tier 1)	Trust Score 70/100, LOW RISK; Eligible limit Rs 15,000 (Tier 1)	**Pass**	Positive
2	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 15000.01	Amount exceeds Tier 1 limit of Rs 15,000	Amount exceeds Tier 1 limit of Rs 15,000	**Pass**	BVA - just above the tier limit
3	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 15000	Loan application #1 submitted. Status: PENDING.	Loan application #1 submitted. Status: PENDING.	**Pass**	BVA - on the tier limit
4	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 5000	You already have an active loan	You already have an active loan	**Pass**	Negative - one active loan only
5	**Check post-condition 1**	-	1 loan stored: #1 Rs 15000, Tier 1, 12 months, 12% p.a., Trust Score 70, PENDING	1 loan stored: #1 Rs 15000, Tier 1, 12 months, 12% p.a., Trust Score 70, PENDING	**Pass**	

Post-conditions
1. Application #1 is stored as PENDING with its tier, tenure, interest rate and Trust Score

Test Case #:	TC-LOAN-02	**Test Case Name:**	Loan amount - invalid values	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-LN-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test that zero, negative, non-numeric and special values are rejected as the loan amount.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His profile shows 5 years of vending and he logged a sale on each of the last 30 days, so his Trust Score is 40 + 20 + 0 + 10 = 70 (LOW RISK)
He has no loan

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 0	Loan amount must be greater than 0	Loan amount must be greater than 0	**Pass**	BVA - zero
2	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = -5000	Loan amount must be greater than 0	Loan amount must be greater than 0	**Pass**	Negative (EP) - negative
3	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = abc	Loan amount must be a number	Loan amount must be a number	**Pass**	Negative (EP) - text
4	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = (empty)	Loan amount must be a number	Loan amount must be a number	**Pass**	Negative (EP) - empty
5	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = nan	Loan amount must be a number	**Run 1:** HTTP 500 Internal Server Error (IntegrityError: NOT NULL constraint failed: loans.amount) **Run 2 (after fix):** Loan amount must be a number	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - special value Run 1 FAILED: defect DEF-01 logged; fixed and re-tested
6	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = inf	Loan amount must be a number	**Run 1:** Amount exceeds Tier 1 limit of Rs 15,000 **Run 2 (after fix):** Loan amount must be a number	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - special value Run 1 FAILED: defect DEF-01 logged; fixed and re-tested
7	**Check post-condition 1**	-	0 loans stored	0 loans stored	**Pass**	

Post-conditions
1. No loan application is stored

Test Case #:	TC-LOAN-03	**Test Case Name:**	Vendor who is not eligible applies	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-TS-5, FR-VEN-6, BR-4
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that a vendor with a Trust Score below 40, or without a profile, cannot apply.				

Pre-conditions
Vendor Sunita Devi (9812345670 / sunita123): profile with 0 years of vending, no sales, Trust Score 20
Vendor Mohan Lal (9898989898 / mohan123) is registered but has NOT saved a profile

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Log in as Sunita. In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 10000	Trust score 20 is below 40. Not eligible	Trust score 20 is below 40. Not eligible	**Pass**	Negative - score below 40
2	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 20000 (also above the Tier-1 limit)	Trust score 20 is below 40. Not eligible	Trust score 20 is below 40. Not eligible	**Pass**	Combination - eligibility is checked first
3	Log in as Mohan. In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 10000	Complete your vendor profile first	Complete your vendor profile first	**Pass**	Negative - no profile
4	**Check post-condition 1**	-	0 loans stored	0 loans stored	**Pass**	

Post-conditions
1. No loan application is stored

Test Case #:	TC-LOAN-04	**Test Case Name:**	Bank officer approves and disburses a loan	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-BNK-2, FR-BNK-5, FR-BNK-6
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test approval and disbursal, and that the loan cannot skip or repeat a state.				

Pre-conditions
Ramesh's application #1 for Rs 15,000 (Tier 1) is PENDING
Bank officer Anil Sharma (9123456780 / bank123) is logged in

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Send a Disburse request for #1 (the button appears only after approval)	Loan #1	Only APPROVED loans can be disbursed	Only APPROVED loans can be disbursed	**Pass**	Negative - invalid state change
2	Click 'Approve' on application #1	Decision = Approve, Remarks = (empty)	Loan #1 APPROVED.	Loan #1 APPROVED.	**Pass**	Positive
3	Send an Approve request for #1 again	Decision = Approve	Only PENDING applications can be reviewed	Only PENDING applications can be reviewed	**Pass**	Negative - invalid state change
4	Click 'Disburse' on loan #1	Loan #1	Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	**Pass**	Positive - EMI from hand calculation
5	**Check post-condition 1**	-	Loan #1 DISBURSED on 04-11-2026; 12 EMIs created	Loan #1 DISBURSED on 04-11-2026; 12 EMIs created	**Pass**	

Post-conditions
1. Loan #1 is DISBURSED with today's date and 12 EMIs are created

Test Case #:	TC-LOAN-05	**Test Case Name:**	Officer rejects; vendor re-applies and withdraws	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-BNK-2, FR-BNK-3, FR-LN-5, FR-LN-6, BR-5
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test rejection rules (remarks required, valid decision, existing loan), that the vendor sees the remarks, and withdrawal.				

Pre-conditions
Ramesh's application #1 for Rs 15,000 is PENDING; Ramesh's Trust Score is 70
Bank officer Anil Sharma (9123456780 / bank123) is logged in

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Leave Remarks empty and click 'Reject' on #1	Decision = Reject, Remarks = (empty)	Remarks are required when rejecting	Remarks are required when rejecting	**Pass**	Negative - remarks required
2	Send the review form with a decision that is not offered	Decision = DELETE, Remarks = x	Decision must be APPROVE or REJECT	Decision must be APPROVE or REJECT	**Pass**	Security - forged value
3	Send an Approve request for application #99	Loan #99 (does not exist)	Loan not found	Loan not found	**Pass**	Negative - unknown loan
4	Enter remarks and click 'Reject' on #1	Decision = Reject, Remarks = Sales record too short	Loan #1 REJECTED.	Loan #1 REJECTED.	**Pass**	Positive
5	Log in as Ramesh and open loan #1	9876543210 / ramesh123	Loan #1 page is shown; Officer remarks: Sales record too short	Loan #1 page is shown; Officer remarks: Sales record too short	**Pass**	Remarks shown to the vendor
6	In 'Micro-credit' enter the amount and click 'Apply for loan'	Amount = 12000	Loan application #2 submitted. Status: PENDING.	Loan application #2 submitted. Status: PENDING.	**Pass**	Positive - may re-apply after rejection
7	Click 'Withdraw' on application #2	Loan #2	Application #2 withdrawn.	Application #2 withdrawn.	**Pass**	Positive
8	**Check post-condition 1**	-	Loan #1 REJECTED (remarks: Sales record too short); Loan #2 WITHDRAWN	Loan #1 REJECTED (remarks: Sales record too short); Loan #2 WITHDRAWN	**Pass**	

Post-conditions
1. Application #1 is REJECTED with the officer's remarks and application #2 is WITHDRAWN

Test Case #:	TC-LOAN-06	**Test Case Name:**	Access control on officer and loan pages	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-AUTH-7, FR-BNK-4, NFR-SEC-7, BR-1
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test that each role can open only the pages allowed to it.				

Pre-conditions
Ramesh's application #1 exists
Registered: Sunita Devi (vendor, 9812345670 / sunita123), Priya Mehta (customer, 9988776655 / priya123), R K Verma (municipal officer, 9090909090 / muni123), Anil Sharma (bank officer, 9123456780 / bank123)

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Log in as Ramesh; type /officer in the browser	URL = /officer	You are not allowed to open that page.	You are not allowed to open that page.	**Pass**	Security - vendor on officer page
2	Log in as Priya (customer); type /loan/1	URL = /loan/1	You are not allowed to open that page.	**Run 1:** Loan #1 page is shown **Run 2 (after fix):** You are not allowed to open that page.	**Fail (Run 1)** Pass (Run 2)	Security - customer Run 1 FAILED: defect DEF-05 logged; fixed and re-tested
3	Log in as R K Verma (municipal officer); type /loan/1	URL = /loan/1	You are not allowed to open that page.	**Run 1:** Loan #1 page is shown **Run 2 (after fix):** You are not allowed to open that page.	**Fail (Run 1)** Pass (Run 2)	Security - municipal officer Run 1 FAILED: defect DEF-05 logged; fixed and re-tested
4	Log in as Sunita (another vendor); type /loan/1	URL = /loan/1	You can only view your own loans.	You can only view your own loans.	**Pass**	Security - other vendor
5	Log in as Anil (bank officer); open /loan/1	URL = /loan/1	Loan #1 page is shown	Loan #1 page is shown	**Pass**	Positive - bank officer
6	Log in as Ramesh; open /loan/1	URL = /loan/1	Loan #1 page is shown	Loan #1 page is shown	**Pass**	Positive - owner

Post-conditions
1. Loan data is visible only to its vendor and to bank officers; nothing is changed

6. EMI Schedule & Repayment (TC-EMI-01 to TC-EMI-06)

Test Case #:	TC-EMI-01	**Test Case Name:**	View the EMI schedule	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-BNK-6, FR-EMI-1
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test the generated reducing-balance EMI schedule against a hand calculation.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months was approved and disbursed today (04-11-2026) by bank officer Anil Sharma
Hand calculation: r = 12% / 12 = 0.01; EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Read the 'Next EMI' card on the dashboard	-	Next EMI Rs 1,332.73 due 2026-12-04	Next EMI Rs 1,332.73 due 2026-12-04	**Pass**	Positive
2	Click 'View' on loan #1 and read the EMI schedule	Loan #1	12 installments; EMI #1 Rs 1,332.73 due 2026-12-04; EMI #12 due 2027-11-04	12 installments; EMI #1 Rs 1,332.73 due 2026-12-04; EMI #12 due 2027-11-04	**Pass**	Monthly due dates
3	Read the principal / interest split of EMI #1	-	EMI #1 = Principal Rs 1,182.73 + Interest Rs 150.00	EMI #1 = Principal Rs 1,182.73 + Interest Rs 150.00	**Pass**	Interest = 15000 x 0.01 = 150
4	Read the split of EMI #2	-	EMI #2 = Principal Rs 1,194.56 + Interest Rs 138.17	EMI #2 = Principal Rs 1,194.56 + Interest Rs 138.17	**Pass**	Interest = 13817.27 x 0.01 (reducing balance)
5	Read the summary cards on the loan page	-	Rs 15,000, Tier 1; 12 months @ 12.0% p.a.; EMIs paid 0/12	Rs 15,000, Tier 1; 12 months @ 12.0% p.a.; EMIs paid 0/12	**Pass**	Positive

Post-conditions
1. The schedule is displayed; nothing is changed

Test Case #:	TC-EMI-02	**Test Case Name:**	Pay EMIs in order	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-EMI-2, FR-EMI-4, FR-EMI-5
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test paying EMIs, and that an installment cannot be paid twice or out of order.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months was approved and disbursed today (04-11-2026) by bank officer Anil Sharma
Hand calculation: r = 12% / 12 = 0.01; EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73
No EMI has been paid

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	On loan #1 enter the amount and click 'Pay' on the installment	Installment #1, Amount = 1332.73	EMI #1 paid on time.	EMI #1 paid on time.	**Pass**	Positive (happy path)
2	Send a payment request for #1 again	Installment #1, Amount = 1332.73	Installment already paid	Installment already paid	**Pass**	Security - forged request
3	Send a payment request for #3 (the page offers only #2)	Installment #3, Amount = 1332.73	Pay earlier installment #2 first	Pay earlier installment #2 first	**Pass**	Security - out of order
4	On loan #1 enter the amount and click 'Pay' on the installment	Installment #2, Amount = 1332.73	EMI #2 paid on time.	EMI #2 paid on time.	**Pass**	Positive
5	**Check post-condition 1**	-	EMI #1 PAID on time, EMI #2 PAID on time; 10 DUE	EMI #1 PAID on time, EMI #2 PAID on time; 10 DUE	**Pass**	

Post-conditions
1. EMIs #1 and #2 are PAID (on time); the other 10 are still DUE

Test Case #:	TC-EMI-03	**Test Case Name:**	Payment amount - invalid values	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-EMI-3
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test that a payment less than the EMI, or one that is not a number, is rejected.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months was approved and disbursed today (04-11-2026) by bank officer Anil Sharma
Hand calculation: r = 12% / 12 = 0.01; EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73
No EMI has been paid; every step pays installment #1

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	On loan #1 enter the amount and click 'Pay' on the installment	Amount = 1332.72 (1 paisa less)	Amount is less than EMI due of Rs 1332.73	Amount is less than EMI due of Rs 1332.73	**Pass**	BVA - just below the EMI
2	On loan #1 enter the amount and click 'Pay' on the installment	Amount = 0	Amount is less than EMI due of Rs 1332.73	Amount is less than EMI due of Rs 1332.73	**Pass**	BVA - zero
3	On loan #1 enter the amount and click 'Pay' on the installment	Amount = abc	Payment amount must be a number	Payment amount must be a number	**Pass**	Negative (EP) - text
4	On loan #1 enter the amount and click 'Pay' on the installment	Amount = (empty)	Payment amount must be a number	Payment amount must be a number	**Pass**	Negative (EP) - empty
5	On loan #1 enter the amount and click 'Pay' on the installment	Amount = nan	Payment amount must be a number	**Run 1:** EMI #1 paid on time. **Run 2 (after fix):** Payment amount must be a number	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - special value Run 1 FAILED: defect DEF-03 logged; fixed and re-tested
6	On loan #1 enter the amount and click 'Pay' on the installment	Amount = inf	Payment amount must be a number	**Run 1:** Installment already paid **Run 2 (after fix):** Payment amount must be a number	**Fail (Run 1)** Pass (Run 2)	Negative (EP) - special value Run 1 FAILED: defect DEF-03 logged; fixed and re-tested
7	**Check post-condition 1**	-	0 installments PAID; 12 DUE	**Run 1:** EMI #1 PAID on time; 11 DUE **Run 2 (after fix):** 0 installments PAID; 12 DUE	**Fail (Run 1)** Pass (Run 2)	Run 1 FAILED: wrong data stored (defect in the steps above); fixed and re-tested

Post-conditions
1. No installment is marked PAID

Test Case #:	TC-EMI-04	**Test Case Name:**	Full repayment raises the score and unlocks Tier 2	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-EMI-5, FR-EMI-6, FR-LN-3, FR-TS-2
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that on-time payments raise the Trust Score and that paying the last EMI closes the loan and unlocks the next tier (progressive lending).				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months was approved and disbursed today (04-11-2026) by bank officer Anil Sharma
Hand calculation: r = 12% / 12 = 0.01; EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73
No EMI has been paid; Ramesh's Trust Score is 70

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Read the Trust Score card	-	Trust Score 70/100, LOW RISK	Trust Score 70/100, LOW RISK	**Pass**	No EMI due yet: repayment = 20
2	On loan #1 enter the amount and click 'Pay' on the installment; then read the Trust Score	Installment #1, Amount = 1332.73	EMI #1 paid on time. -> Trust Score 90/100, LOW RISK	EMI #1 paid on time. -> Trust Score 90/100, LOW RISK	**Pass**	1 of 1 on time: repayment = 40
3	Pay installments #2 to #12 in order, each with the amount shown	11 payments	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier!	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier!	**Pass**	Positive
4	Open the dashboard and read the eligible-limit card	-	Eligible limit Rs 25,000 (Tier 2)	Eligible limit Rs 25,000 (Tier 2)	**Pass**	Next tier unlocked
5	**Check post-condition 1**	-	Loan #1 CLOSED; 12 of 12 EMIs PAID	Loan #1 CLOSED; 12 of 12 EMIs PAID	**Pass**	

Post-conditions
1. Loan #1 is CLOSED, all 12 EMIs are PAID and Ramesh can now borrow up to Rs 25,000

Test Case #:	TC-EMI-05	**Test Case Name:**	Invalid payment requests	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-EMI-1, FR-BNK-5, NFR-SEC-7
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**Run 1: FAIL → Run 2: PASS**
Short Description:	Test forged payment requests: unknown installment or loan, someone else's loan, a loan not disbursed.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered and logged in
His Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months was approved and disbursed today (04-11-2026) by bank officer Anil Sharma
Hand calculation: r = 12% / 12 = 0.01; EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73
Sunita Devi (9812345670 / sunita123) is a registered vendor
Mohd. Salim (9811122233 / salim123) has loan #2 APPROVED but not yet disbursed

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Send a payment request for installment #13	Loan #1, Installment #13, Amount = 1332.73	Installment not found	Installment not found	**Pass**	Security - no such installment
2	Send a payment request for installment #0	Loan #1, Installment #0, Amount = 1332.73	Installment not found	Installment not found	**Pass**	BVA - below #1
3	Send a payment request for loan #99	Loan #99, Installment #1, Amount = 1332.73	Loan not found	**Run 1:** Loan not found | Loan not found **Run 2 (after fix):** Loan not found	**Fail (Run 1)** Pass (Run 2)	Negative - unknown loan Run 1 FAILED: defect DEF-06 logged; fixed and re-tested
4	Log in as Sunita; send a payment for Ramesh's loan #1	Loan #1, Installment #1	You can only pay your own loans	**Run 1:** You can only pay your own loans | You can only view your own loans. **Run 2 (after fix):** You can only pay your own loans	**Fail (Run 1)** Pass (Run 2)	Security - someone else's loan Run 1 FAILED: defect DEF-06 logged; fixed and re-tested
5	Log in as Salim; send a payment for his loan #2	Loan #2, Installment #1	Loan is not active for repayment	Loan is not active for repayment	**Pass**	Negative - loan not disbursed
6	**Check post-condition 1**	-	0 installments PAID; 12 DUE	0 installments PAID; 12 DUE	**Pass**	

Post-conditions
1. No installment is marked PAID

Test Case #:	TC-EMI-06	**Test Case Name:**	Late payment and default check (90 / 91 days overdue)	**Page:**	1 of 1
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)	**Type:**	System test
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026	**Requirements:**	FR-BNK-7, BR-6, FR-EMI-5, FR-TS-2
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026	**Result:**	**PASS**
Short Description:	Test that a loan becomes DEFAULTED only when an EMI is MORE than 90 days overdue, that it can still be repaid, and that a late payment lowers the Trust Score.				

Pre-conditions
Vendor Ramesh Kumar (9876543210 / ramesh123) is registered
His loan #1 (Rs 15,000) is DISBURSED and no EMI is paid
EMI #1's due date is 06-08-2026 (exactly 90 days ago)
Bank officer Anil Sharma (9123456780 / bank123) is logged in

Step	Action	Input (Test Data)	Expected System Response	Actual Output	Pass/ Fail	Comment
1	Click 'Run default check (90+ days overdue)'	EMI #1 due 06-08-2026 (90 days overdue)	Default check complete: 0 loan(s) marked DEFAULTED.	Default check complete: 0 loan(s) marked DEFAULTED.	**Pass**	BVA - exactly 90 days: not defaulted
2	One day later, click 'Run default check' again	EMI #1 due 05-08-2026 (91 days overdue)	Default check complete: 1 loan(s) marked DEFAULTED.	Default check complete: 1 loan(s) marked DEFAULTED.	**Pass**	BVA - 91 days: defaulted
3	**Check post-condition 1**	-	Loan #1 DEFAULTED	Loan #1 DEFAULTED	**Pass**	
4	Log in as Ramesh. On loan #1 enter the amount and click 'Pay' on the installment	Installment #1, Amount = 1332.73	EMI #1 paid late.	EMI #1 paid late.	**Pass**	A defaulted loan can still be repaid
5	Read the Trust Score card	-	Trust Score 50/100, MEDIUM RISK	Trust Score 50/100, MEDIUM RISK	**Pass**	0 of 1 on time: repayment = 0; 40 + 0 + 0 + 10

Post-conditions
1. Loan #1 is DEFAULTED; EMI #1 is PAID (late)

7. Defect Report - Run 1 (first version) and Re-test - Run 2

Run 1 executed the same test cases on the first version of the code (git commit 0ce1911): 7 test cases FAILED. The real Actual Output of every failed step is shown below. Each failure was logged as a defect, fixed, and the whole suite was executed again (regression testing).

Test case / step	Input (Test Data)	Expected System Response	Actual Output (Run 1)	Pass/ Fail	Actual Output (Run 2, after fix)	Pass/ Fail
TC-AUTH-02 step 2	Name = 12345, Mobile = 9876543211	Name can contain only letters, spaces and dots	Registration successful. Please log in.	**Fail** DEF-04	Name can contain only letters, spaces and dots	**Pass**
TC-AUTH-02 step 3	Name = 'A' x 51 characters, Mobile = 9876543212	Name must be at most 50 characters	Registration successful. Please log in.	**Fail** DEF-04	Name must be at most 50 characters	**Pass**
TC-AUTH-02 step 6	-	2 users stored: 'A' x 50 (VENDOR), Mohd. Salim (VENDOR)	4 users stored: 12345 (VENDOR), 'A' x 51 (VENDOR), 'A' x 50 (VENDOR), Mohd. Salim (VENDOR)	**Fail**	2 users stored: 'A' x 50 (VENDOR), Mohd. Salim (VENDOR)	**Pass**
TC-SALE-03 step 5	Amount = nan	Sale amount must be a number	HTTP 500 Internal Server Error (IntegrityError: NOT NULL constraint failed: sales.amount)	**Fail** DEF-01	Sale amount must be a number	**Pass**
TC-SALE-03 step 6	Amount = inf	Sale amount must be a number	Sale amount cannot exceed Rs 1,00,000 per day	**Fail** DEF-01	Sale amount must be a number	**Pass**
TC-SALE-05 step 3	Date = 05-08-2026 (91 days ago)	Sale date cannot be more than 90 days old	Sale logged. Your trust score has been updated.	**Fail** DEF-02	Sale date cannot be more than 90 days old	**Pass**
TC-SALE-05 step 7	-	1 sale stored: 06-08-2026 Rs 1500 UPI	2 sales stored: 06-08-2026 Rs 1500 UPI, 05-08-2026 Rs 1500 UPI	**Fail**	1 sale stored: 06-08-2026 Rs 1500 UPI	**Pass**
TC-LOAN-02 step 5	Amount = nan	Loan amount must be a number	HTTP 500 Internal Server Error (IntegrityError: NOT NULL constraint failed: loans.amount)	**Fail** DEF-01	Loan amount must be a number	**Pass**
TC-LOAN-02 step 6	Amount = inf	Loan amount must be a number	Amount exceeds Tier 1 limit of Rs 15,000	**Fail** DEF-01	Loan amount must be a number	**Pass**
TC-LOAN-06 step 2	URL = /loan/1	You are not allowed to open that page.	Loan #1 page is shown	**Fail** DEF-05	You are not allowed to open that page.	**Pass**
TC-LOAN-06 step 3	URL = /loan/1	You are not allowed to open that page.	Loan #1 page is shown	**Fail** DEF-05	You are not allowed to open that page.	**Pass**
TC-EMI-03 step 5	Amount = nan	Payment amount must be a number	EMI #1 paid on time.	**Fail** DEF-03	Payment amount must be a number	**Pass**
TC-EMI-03 step 6	Amount = inf	Payment amount must be a number	Installment already paid	**Fail** DEF-03	Payment amount must be a number	**Pass**
TC-EMI-03 step 7	-	0 installments PAID; 12 DUE	EMI #1 PAID on time; 11 DUE	**Fail**	0 installments PAID; 12 DUE	**Pass**
TC-EMI-05 step 3	Loan #99, Installment #1, Amount = 1332.73	Loan not found	Loan not found | Loan not found	**Fail** DEF-06	Loan not found	**Pass**
TC-EMI-05 step 4	Loan #1, Installment #1	You can only pay your own loans	You can only pay your own loans | You can only view your own loans.	**Fail** DEF-06	You can only pay your own loans	**Pass**

A 'Check post-condition' step fails in Run 1 when invalid input was stored in the database; it has no defect of its own - it is caused by the defect in the steps above it.

Defect log

Defect	Severity	Found by	What happened (Run 1)	Cause	Fix	Re-test (Run 2)
DEF-01	High	TC-SALE-03 step 5 TC-SALE-03 step 6 TC-LOAN-02 step 5 TC-LOAN-02 step 6	Typing 'nan' as a sale or loan amount crashed the app (HTTP 500 Internal Server Error); 'inf' gave a misleading message.	Python's float() accepts 'nan' and 'inf'. NaN passed every comparison and SQLite stored it as NULL in a NOT NULL column.	New helper parse_number() in database.py rejects values that are not finite numbers.	Pass - closed

Defect	Severity	Found by	What happened (Run 1)	Cause	Fix	Re-test (Run 2)
DEF-02	Medium	TC-SALE-05 step 3	A sale dated 91 days ago (or 01-01-1900) was accepted.	Only future dates were checked.	vendor.log_daily_sale() rejects entries more than 90 days old (MAX_BACKDATE_DAYS); FR-VEN-5 updated.	Pass - closed
DEF-03	High	TC-EMI-03 step 5 TC-EMI-03 step 6	Paying an EMI with amount 'nan' marked the installment PAID.	Same as DEF-01: the 'less than EMI' check is False for NaN.	loan.pay_emi() uses parse_number().	Pass - closed
DEF-04	Low	TC-AUTH-02 step 2 TC-AUTH-02 step 3	Names such as '12345' or a 51-character string were accepted at registration.	The name was only checked for being non-empty.	auth.validate_name(): letters, spaces, dots and apostrophes only, at most 50 characters; FR-AUTH-1 updated.	Pass - closed
DEF-05	High	TC-LOAN-06 step 2 TC-LOAN-06 step 3	A customer or municipal officer could open any vendor's loan page and see the amount and EMIs.	The /loan/<id> page only required a logged-in user and checked ownership for vendors only.	Only the owning vendor and bank officers may open a loan page (LOAN_VIEW_ROLES in app.py).	Pass - closed
DEF-06	Low	TC-EMI-05 step 3 TC-EMI-05 step 4	Paying an EMI of a loan that does not exist, or of another vendor's loan, showed two error messages ('Loan not found | Loan not found').	The error page redirected to the loan page, which reported a second error.	The pay_emi route redirects to the dashboard when the loan is missing or not the vendor's own.	Pass - closed

8. Combination Testing (supporting the test cases)

In addition to the documented test cases, every form was submitted with every combination (cartesian product) of the valid, invalid and boundary values below. An oracle written from the SRS rules (tests/combinations.py) gives the message each combination must show; a combination passes only if the message matches, the app does not crash and rejected input leaves the database unchanged.

Form	Values tried for each field	Combinations	Run 1 (first version)	Run 2 (after fixes)
Register	**Name:** 'Ramesh Kumar', 'Mohd. Salim', '', '12345', 'A' x 51 **Mobile:** '9876543210', '987654321', '98765432101', '5876543210', '98765abcde', '' **Password:** 'ramesh123', 'ab123', 'abcdefgh', '12345678', '' **Confirm:** 'same', 'different' **Role:** 'VENDOR', 'BANK_OFFICER', '', 'ADMIN'	1,200	960 correct, 0 crashed	1,200 correct, 0 crashed
Log in	**Mobile:** '9876543210', ' 9876543210 ', '9000000001', '', "' OR '1'='1" **Password:** 'ramesh123', 'RAMESH123', 'ramesh124', '', "' OR '1'='1"	25	25 correct, 0 crashed	25 correct, 0 crashed
Business profile	**Business:** 'Fruits', '', ' ' **Zone:** 'Sector 14', '' **Years:** '0', '5', '60', '61', '-1', 'abc', '', '2.5'	48	48 correct, 0 crashed	48 correct, 0 crashed
Daily sales	**Date:** '2026-11-04', '2026-11-03', '2026-08-06', '2026-08-05', '2026-11-05', '2026-02-30', '', 'abc' **Amount:** '1500', '1', ' 250 ', '1e3', '100000', '100000.01', '0', '-5', 'abc', '', 'nan', 'inf' **Mode:** 'UPI', 'CASH', 'upi', 'CARD', ''	480	405 correct, 8 crashed	480 correct, 0 crashed
Loan application	**Vendor:** 'eligible (score 70)', 'score 20', 'no profile', 'has a pending loan' **Amount:** '15000', '1', '15000.01', '0', '-1', 'abc', '', 'nan', 'inf'	36	28 correct, 1 crashed	36 correct, 0 crashed
EMI payment	**Installment:** 1, 2, 12, 13, 0 **Amount:** '1332.73', '1332.72', '5000', '0', '-1', 'abc', '', 'nan', 'inf'	45	43 correct, 0 crashed	45 correct, 0 crashed
Role x page access	**Role:** 'Guest (not logged in)', 'Owner vendor (Ramesh)', 'Other vendor (Sunita)', 'Bank officer', 'Customer', 'Municipal officer' **Request:** 'GET /dashboard', 'GET /officer', 'GET /loan/1', 'POST /vendor/sales', 'POST /vendor/profile', 'POST /vendor/loan/apply', 'POST /vendor/loan/1/pay/1', 'POST /officer/loan/1/review', 'POST /officer/loan/1/disburse', 'POST /officer/mark-defaults'	60	58 correct, 0 crashed	60 correct, 0 crashed
Total		**1,894**	**1,567 correct**	**1,894 correct**

9. Requirements Traceability Matrix

Each SRS requirement covered by the implemented modules, and the test cases that verify it.

Requirement	Verified by test case(s)	Status
FR-AUTH-1	TC-AUTH-01, TC-AUTH-02, TC-AUTH-03, TC-AUTH-04	Covered - Pass
FR-AUTH-2	TC-AUTH-01, TC-AUTH-03	Covered - Pass
FR-AUTH-3	TC-AUTH-04	Covered - Pass
FR-AUTH-4	TC-AUTH-01	Covered - Pass
FR-AUTH-5	TC-AUTH-05	Covered - Pass
FR-AUTH-6	TC-AUTH-06	Covered - Pass
FR-AUTH-7	TC-AUTH-05, TC-LOAN-06	Covered - Pass
FR-VEN-1	TC-SALE-01, TC-SALE-02	Covered - Pass
FR-VEN-2	TC-SALE-02	Covered - Pass
FR-VEN-3	TC-SALE-01, TC-SALE-05	Covered - Pass
FR-VEN-4	TC-SALE-03, TC-SALE-04, TC-SALE-05	Covered - Pass
FR-VEN-5	TC-SALE-01, TC-SALE-05	Covered - Pass
FR-VEN-6	TC-SALE-01, TC-LOAN-03	Covered - Pass
FR-VEN-7	TC-SALE-06	Covered - Pass
FR-TS-1	TC-SALE-06, TC-TS-01, TC-TS-06	Covered - Pass
FR-TS-2	TC-TS-02, TC-TS-03, TC-TS-04, TC-TS-05, TC-EMI-04, TC-EMI-06	Covered - Pass
FR-TS-3	TC-TS-01, TC-TS-04	Covered - Pass
FR-TS-4	TC-TS-01, TC-TS-02, TC-TS-03, TC-TS-04	Covered - Pass
FR-TS-5	TC-TS-01, TC-TS-02, TC-LOAN-03	Covered - Pass
FR-TS-6	TC-TS-01	Covered - Pass
FR-LN-1	TC-LOAN-01	Covered - Pass
FR-LN-2	TC-LOAN-01	Covered - Pass
FR-LN-3	TC-EMI-04	Covered - Pass
FR-LN-4	TC-LOAN-01, TC-LOAN-02	Covered - Pass
FR-LN-5	TC-LOAN-01, TC-LOAN-05	Covered - Pass
FR-LN-6	TC-LOAN-01, TC-LOAN-05	Covered - Pass
FR-BNK-2	TC-LOAN-04, TC-LOAN-05	Covered - Pass
FR-BNK-3	TC-LOAN-05	Covered - Pass
FR-BNK-4	TC-LOAN-06	Covered - Pass
FR-BNK-5	TC-LOAN-04, TC-EMI-05	Covered - Pass
FR-BNK-6	TC-LOAN-04, TC-EMI-01	Covered - Pass
FR-BNK-7	TC-EMI-06	Covered - Pass
FR-EMI-1	TC-EMI-01, TC-EMI-05	Covered - Pass
FR-EMI-2	TC-EMI-02	Covered - Pass
FR-EMI-3	TC-EMI-03	Covered - Pass
FR-EMI-4	TC-EMI-02	Covered - Pass
FR-EMI-5	TC-EMI-02, TC-EMI-04, TC-EMI-06	Covered - Pass
FR-EMI-6	TC-EMI-04	Covered - Pass
BR-1	TC-LOAN-06	Covered - Pass

Requirement	Verified by test case(s)	Status
BR-4	TC-LOAN-03	Covered - Pass
BR-5	TC-LOAN-05	Covered - Pass
BR-6	TC-EMI-06	Covered - Pass
BR-8	TC-TS-05	Covered - Pass
NFR-SEC-1	TC-AUTH-01	Covered - Pass
NFR-SEC-2	TC-AUTH-06	Covered - Pass
NFR-SEC-4	TC-AUTH-05	Covered - Pass
NFR-SEC-7	TC-LOAN-06, TC-EMI-05	Covered - Pass

10. How to Re-execute the Tests

```
cd C:\Users\nancy\OneDrive\Desktop\SE_Project
python tests/run_test_cases.py --module AUTH --details   (one module, every step's input / expected / actual)
python tests/run_test_cases.py --before-fix              (Run 1: first version - shows the FAILED steps)
python tests/run_test_cases.py                           (Run 2: all test cases + combination testing)
python -m pytest tests -v                                (the same test cases under pytest)
python docs/build_test_cases.py                          (regenerates this document)
```